

MarketMind AI

Small Business Sales Intelligence & Analytics Platform

React 18 | FastAPI | Neon PostgreSQL | XGBoost | scikit-learn

Project Report

Team 2 - Infosys Springboard Capstone

Neelam Rishika Damini (Damini2006)

Generated: September 09, 2026

GitHub: [springboardmentor24052s-tech/Team_2_Small_Biz_AI](https://github.com/springboardmentor24052s-tech/Team_2_Small_Biz_AI)

Table of Contents

1. [Project Overview](#)
2. [Architecture & Technology Stack](#)
3. [Features & Modules](#)
4. [Database Schema \(28 Tables\)](#)
5. [API Endpoints](#)
6. [Role-Based Access Control](#)
7. [Machine Learning & AI Pipeline](#)
8. [Frontend - Pages & Components](#)
9. [Security](#)
10. [Performance](#)
11. [Milestone Timeline & Work Report](#)
12. [Bug Fix Log \(16 Bugs\)](#)
13. [Key Learnings](#)
14. [Getting Started](#)

1. Project Overview

MarketMind AI is a full-stack SaaS application built to give small retail businesses access to enterprise-grade analytics. Designed specifically for Indian small businesses, the platform delivers real-time insights across sales, inventory, customers, and finances - with ML models for forecasting, anomaly detection, and personalized recommendations.

Highlights

28+	Pages covering sales, inventory, customers, invoices, analytics, and AI
5	Languages - English, Hindi, Telugu, Tamil, Kannada (full app translation)
4	Roles - Business Owner, Store Manager, Sales Executive, Admin (RBAC)
28	PostgreSQL tables on Neon (serverless, connection pooling, SSL)
PWA	Installable on mobile and desktop with offline caching
WS	Real-time WebSocket-powered live alerts and notifications
87	Git commits across 4 milestones (Jul 24 - Sep 5, 2026)

2. Architecture & Technology Stack

System Architecture

Three-tier architecture: React frontend communicates with FastAPI backend via REST API and WebSocket, backed by Neon PostgreSQL with 28 tables and connection pooling.

Frontend (React)			Backend (FastAPI)			Neon PostgreSQL		
Dashboard	Sales		Auth Router					
Inventory	Customers	<->	Sales / Inventory Router	<->		28 Tables		
AI/ML	Reports	API	Analytics / AI Engine	ORM		Connection Pool		
Builder	Settings		WebSocket Alert Handler			SSL + Auto-scale		

Frontend Stack

Technology	Purpose
React 18	UI framework with lazy-loaded routes and code splitting
Vite	Build tool with HMR and optimized bundling (~590 KB gzipped ~178 KB)
Tailwind CSS	Utility-first styling with dark mode support
React Router v6	Client-side routing with role-based guards
Recharts	Interactive data visualizations and charts
react-grid-layout	Drag-and-drop dashboard builder with 7 widget types
react-i18next	Internationalization - 5 languages, centralized i18n.js
jspdf + html2canvas	PDF export from any page
xlsx	Excel spreadsheet export
qrcode.react	QR code generation for products and invoices

Backend Stack

Technology	Purpose
FastAPI	Async Python web framework with automatic OpenAPI docs
SQLAlchemy	ORM with connection pooling and query optimization
Pydantic v2	Data validation and serialization with strict typing
python-jose	JWT token generation and validation (12-hour expiry)
bcrypt	Password hashing with salt
scikit-learn	K-Means clustering, Isolation Forest, Logistic Regression
XGBoost	Sales forecasting model (14/30-day predictions)

WebSocket	Real-time alert broadcasting to all connected clients
Neon PostgreSQL	Serverless cloud database with pooler endpoint and SSL

3. Features & Modules

Core Business Modules

Module	Description
Dashboard	Role-based KPIs, revenue trends, top customers, animated count-up
Sales	Record, track & analyze revenue; CSV upload; server-side pagination
Inventory	Real-time stock tracking, low-stock alerts, reorder recommendations
Invoices	Professional invoice generation with QR codes and status tracking
Customers	RFM segmentation, CLV prediction, churn risk flags
Categories	Product organization with hierarchy management
Suppliers	Supplier CRUD with contact and catalog tracking
Team	Team member management with role assignment

AI & Machine Learning

Feature	Description
Sales Forecasting	XGBoost-based 14/30-day revenue predictions with confidence bands
Customer Segmentation	K-Means clustering by Recency, Frequency, Monetary value (RFM)
Churn Prediction	Logistic Regression with StratifiedKFold cross-validation
Anomaly Detection	Statistical outlier detection (2.5 sigma) + business rule anomalies
Product Recommendations	Collaborative filtering + association rule mining
Revenue Prediction	Category-based revenue forecasting with seasonality adjustment
CLV Prediction	Customer Lifetime Value scoring for prioritization
AI Chatbot	RAG knowledge base with natural language replies and TTS

Analytics & Reporting

Feature	Description
Report Templates	6 pre-built templates: Sales, Inventory, Customer, Financial, Anomaly, AI
Scheduled Reports	Daily/weekly/monthly automated report delivery with history log
Dashboard Builder	Drag-and-drop customizable dashboard with 7 widget types + 3 templates
Funnel Analysis	Customer journey tracking with conversion rates at each stage
Comparison	Side-by-side period-over-period analytics
Export PDF/Excel	One-click export from any page using jsPDF and xlsx
Activity Log	Paginated audit trail with IP address logging

Platform Features

Feature	Description
Multi-Language i18n	English, Hindi, Telugu, Tamil, Kannada - full app translation
Dark/Light Mode	System-aware theme with manual toggle in settings
PWA	Installable on mobile/desktop; offline caching via service worker
Real-Time Alerts	WebSocket live notifications with browser push support
Command Palette	Ctrl+K quick navigation across all pages
Audit Trail	IP-logged action tracking stored in Neon audit_logs table
TOTP 2FA	Authenticator app support with QR code setup and 10 backup codes
Onboarding Wizard	5-step guided setup flow for new users
QR Code Generator	Product and invoice QR codes via qrcode.react
Error Boundary	Graceful error handling with user-friendly fallback UI

4. Database Schema (28 Tables - Neon PostgreSQL)

MarketMind AI uses 28 PostgreSQL tables on Neon. All foreign key columns are indexed for query performance. Connection pooling is configured via the Neon pooler endpoint with SSL.

Core Tables (11)

Table Name	Description
users	User accounts with role, hashed_password, 2FA, business link
businesses	Business profiles with owner, name, category, location
customers	Customer records with RFM scores and CLV data
products	Product catalog with category, supplier, price, stock
sales	Sales transactions with business_id isolation
sale_items	Line items linking sales to products with qty & price
invoices	Invoice records with QR code and status tracking
categories	Product category hierarchy
suppliers	Supplier profiles with contact and catalog info
inventory	Real-time stock levels per product
inventory_transactions	Stock movement audit trail

AI/ML Tables (4)

Table Name	Description
forecasts	Stored XGBoost forecast results per business
churn_predictions	Logistic Regression churn scores per customer
product_recommendations	Collaborative filtering output per customer
customer_segments	K-Means cluster assignments with RFM labels

Analytics Tables (3)

Table Name	Description
anomaly_alerts	Detected anomalies with severity, category, score
inventory_alerts	Low-stock and reorder alerts with product details
notifications	General notification queue with read/unread state

Platform Tables (10)

Table Name	Description
audit_logs	IP-logged audit trail for all user actions

dashboard_layouts	Saved drag-and-drop dashboard configurations
custom_report_templates	User-defined report template configurations
scheduled_reports	Report delivery schedules with frequency and history
prediction_history	Historical ML prediction results for trend analysis
chat_history	AI chatbot conversation logs per user
uploaded_datasets	User-uploaded CSV/Excel dataset metadata
alembic_version	Database migration version tracking

5. API Endpoints

Full interactive API documentation at <http://localhost:8000/docs> (Swagger UI). The backend has 19 router files covering all features.

Authentication

POST	/api/auth/register	Create new account
POST	/api/auth/login	Login and receive JWT token
POST	/api/auth/send-otp	Send password reset OTP email
POST	/api/auth/reset-password-otp	Reset password with OTP
GET	/api/auth/me	Get current user profile
PUT	/api/auth/profile	Update profile fields
PUT	/api/auth/change-password	Change password with verification

Business Operations

GET/POST	/api/sales/	List/create sales (paginated 50/page)
GET/POST	/api/inventory/products	List/create products
GET/POST	/api/customers/	List/create customers
GET/POST	/api/invoices/	List/create invoices
GET/POST	/api/categories/	List/create categories
GET/POST	/api/suppliers/	List/create suppliers
GET	/api/inventory/alerts	Low-stock alerts with product details

AI & Analytics

GET	/api/ai/forecast	Sales forecasting (14/30-day XGBoost)
GET	/api/ai/segmentation	Customer segmentation (K-Means RFM)
GET	/api/ai/churn	Churn predictions (Logistic Regression)
GET	/api/ai/anomalies	Anomaly detection (stats + business rules)
GET	/api/ai/recommendations	Product recommendations (collab filtering)
GET	/api/ai/clv	Customer Lifetime Value predictions
GET	/api/analytics/kpis	Dashboard KPIs with business_id isolation
GET	/api/analytics/charts	Chart data for revenue, sales, inventory
GET	/api/revenue-prediction	Category-based revenue forecasting

Platform

GET	/api/audit/logs	Paginated audit trail with IP
WS	/ws/alerts/{business_id}	Real-time WebSocket alert stream
GET/POST	/api/user-data/dashboard-layouts	CRUD for dashboard layouts
GET/POST	/api/user-data/report-templates	CRUD for report templates
GET/POST	/api/user-data/scheduled-reports	CRUD for scheduled reports

GET	/api/activity/recent	Recent activity feed
GET	/api/notifications/unread-count	Unread notification count

6. Role-Based Access Control (RBAC)

All pages and API endpoints are protected by role-based guards. The ProtectedRoute component in React checks user.role against an allowedRoles array. Backend endpoints use dependency-injected role checks.

Feature	Bus. Owner	Store Mgr	Sales Exec	Admin
Dashboard	Yes	Yes	Yes	Yes
Sales	Yes	Yes	Yes	Yes
Inventory	Yes	Yes	Yes	Yes
Invoices	Yes	Yes	Yes	Yes
Customers	Yes	Yes	Yes	Yes
Categories	Yes	Yes	No	Yes
Suppliers	Yes	Yes	No	Yes
Team	Yes	No	No	Yes
Datasets	Yes	No	No	Yes
Forecasting	Yes	Yes	No	Yes
Segmentation	Yes	Yes	Yes	Yes
Churn Prediction	Yes	Yes	No	Yes
Recommendations	Yes	Yes	Yes	Yes
Anomalies	Yes	Yes	No	Yes
Revenue Predict.	Yes	Yes	No	Yes
Dashboard Builder	Yes	No	No	Yes
Report Templates	Yes	No	No	Yes
Settings	Yes	Yes	Yes	Yes
Audit Trail	Yes	No	No	Yes

7. Machine Learning & AI Pipeline

ML Modules

anomaly_detection.py

Statistical outlier detection using Z-score (2.5 sigma threshold) across sales amount, quantity, and frequency. Generates AnomalyAlert records with severity scoring.

business_rule_anomalies.py

Rule-based anomaly detection: zero-price sales, bulk orders from infrequent customers, unusually high quantities, duplicate transactions within short windows.

churn.py

Logistic Regression classifier with StratifiedKFold cross-validation. Features: recency, frequency, monetary, avg order value. Outputs probability score per customer.

recommendation_engine.py

Collaborative filtering using co-purchase matrix + association rule mining. Generates top-N product recommendations per customer based on purchase history.

recommendations.py

Higher-level recommendation orchestration layer integrating with the Neon product_recommendations table and exposing REST endpoints.

inference.py

Unified ML inference entry point - loads trained models (XGBoost, K-Means) from disk and runs predictions on incoming request data.

Trained Models

Trained model artifacts are persisted in backend/app/trained_models/ and loaded at startup via the warm-up task in main.py. Models are retrained when new data is uploaded via the Datasets page.

Model	Task	Details
XGBoost Regressor	Sales Forecasting	14/30-day revenue predictions with confidence bounds
K-Means Clustering	Customer Segmentation	RFM-based clusters: High Value, Medium, Low, At-Risk
Logistic Regression	Churn Prediction	Binary classifier with StratifiedKFold cross-validation
Isolation Forest	Anomaly Detection	Unsupervised outlier detection (scikit-learn)
Collab. Filtering	Recommendations	Co-purchase matrix with association rule scoring

8. Frontend - Pages & Components

Pages (27 Total)

- LandingPage.jsx - Animated 3D landing page with Three.js interactive product preview
- Login.jsx - JWT authentication with role selection and error handling
- Register.jsx - Account creation with role, business name, and password strength
- ForgotPassword.jsx - OTP-based password reset flow (email -> OTP -> new password)
- Dashboard.jsx - Role-based KPIs with animated count-up and Recharts visualizations
- Sales.jsx - Sales CRUD with CSV upload, server-side pagination (50/page)
- Inventory.jsx - Product & stock management with low-stock alerts
- Customers.jsx - Customer list with RFM scores, CLV, and churn flags
- Invoices.jsx - Invoice generation with QR code and status tracking
- Categories.jsx - Category CRUD with hierarchy management
- Suppliers.jsx - Supplier management with contact details
- Team.jsx - Team member management with role assignment
- Forecasting.jsx - XGBoost sales forecast charts (14/30-day)
- Segmentation.jsx - K-Means RFM segmentation with cluster visualization
- Churn.jsx - Churn prediction table with risk scores and model metrics
- Recommendations.jsx - Product recommendation cards per customer
- Anomalies.jsx - Anomaly dashboard with severity heatmap and top entities
- RevenuePrediction.jsx - Category-based revenue forecasting with seasonality
- Comparison.jsx - Side-by-side period-over-period analytics
- FunnelAnalysis.jsx - Customer journey funnel with conversion rate metrics
- DashboardBuilder.jsx - Drag-and-drop dashboard builder with 7 widget types
- ReportTemplates.jsx - 6 pre-built report templates with PDF/Excel export
- ScheduledReports.jsx - Report delivery scheduling with CRUD and history
- ActivityLog.jsx - Paginated activity feed from audit_logs
- AuditTrail.jsx - IP-logged audit trail with filters
- Datasets.jsx - Dataset upload management (CSV/Excel)
- Settings.jsx - Profile / Security (2FA, password) / Preferences tabs

Key Components

Component	Purpose
Layout.jsx	Sidebar + header shell with responsive hamburger toggle
ChatBot.jsx	AI chatbot overlay with RAG knowledge base and TTS
LiveAlerts.jsx	WebSocket-powered real-time alert toasts
LanguageSwitcher.jsx	Flag dropdown for 5-language selection
OnboardingWizard.jsx	5-step guided setup wizard for new users

CommandPalette.jsx	Ctrl+K quick navigation search overlay
InstallBanner.jsx	PWA install instructions with platform detection
PageSkeleton.jsx	Consistent skeleton loading placeholders
ProtectedRoute.jsx	Role-based route guard wrapping all pages
ErrorBoundary.jsx	React error boundary with graceful fallback UI

9. Security

Security Feature	Details
JWT Authentication	12-hour token expiry; tokens stored in localStorage with auto-attach interceptors
bcrypt Hashing	Password hashing with salt via passlib[bcrypt]
Rate Limiting	Login: 10 attempts/5min; Register: 5 attempts/5min per IP
RBAC	Role-based access control on all API endpoints and React routes
CORS Protection	Configurable allowed origins via CORS_ORIGINS env variable
OTP Reset	Email-based OTP for password resets; codes expire after 10 minutes
TOTP 2FA	Authenticator app (TOTP) with QR code setup and 10 backup codes
Audit Trail	All user actions logged with IP address in audit_logs table
No Hardcoded Secrets	All sensitive values via environment variables (.env)
Input Validation	Pydantic v2 strict validation on all request bodies

10. Performance

Metric	Value
Initial bundle size	~590 KB (gzipped ~178 KB) with Vite code splitting
Code splitting	27 lazy-loaded route chunks via React.lazy() + Suspense
API caching	TTL-based in-memory cache for KPIs, sales, analytics (120s TTL)
DB indexes	All 31 foreign key columns indexed in Neon PostgreSQL
GZip compression	Enabled for API responses larger than 1 KB
Connection pooling	Neon serverless pooler endpoint with pool_pre_ping=True
Keepalive ping	Background startup task pings DB every 4 minutes to prevent idle drops
Skeleton loading	PageSkeleton placeholders on all pages to prevent layout shift
Pagination	Server-side pagination (50 records/page) for sales and audit logs
Lazy images	All chart data fetched asynchronously with loading states

11. Milestone Timeline & Work Report

Author: Neelam Rishika Damini (Damini2006) | Team 2 | Infosys Springboard Capstone

Milestone 1: Foundation & Core Architecture

Jul 24 - Aug 11, 2026 | 22 commits

- Project initialization with React + Vite frontend and FastAPI backend
- Complete authentication system: login, register, forgot password with OTP
- Role-based access control (Business Owner, Store Manager, Sales Executive, Admin)
- Animated 3D landing page with Three.js interactive product preview
- Core CRUD pages: Sales, Inventory, Customers, Invoices, Categories, Suppliers
- Dashboard with KPIs, revenue trends, and top customer charts
- AI/ML pages: Forecasting, Customer Segmentation, Churn Prediction
- Multi-tenant architecture with business_id isolation on Neon PostgreSQL

Milestone 2: AI/ML Integration & Analytics

Aug 11 - Aug 28, 2026 | 14 commits

- Full AI/ML pipeline: Sales Forecasting (XGBoost), Segmentation (K-Means), Churn (Logistic Regression)
- Anomaly Detection with 2.5 sigma statistical outlier detection and business rule anomalies
- Revenue Prediction with category-based forecasting and seasonality modeling
- Product Recommendations using collaborative filtering + association rule mining
- Funnel Analysis with customer journey tracking and conversion rates
- Comparison page for side-by-side period-over-period analytics
- ChatBot with natural text replies, auto-speak TTS, and RAG knowledge base
- Dashboard Builder with drag-and-drop widgets and guided tour for all 4 roles
- Merged ML code from 3 team members (Pramodh, Pallavi, Kavya)

Milestone 3: Platform Features & PWA

Sep 3 - Sep 4, 2026 | 20 commits

- 5-language i18n support (English, Hindi, Telugu, Tamil, Kannada) with LanguageSwitcher
- PWA setup: manifest.json, service worker, app icons (72-512px), InstallBanner
- WebSocket-powered LiveAlerts with toast notifications and browser push support
- Backend WebSocket endpoint for real-time alert broadcasting
- 6 Report Templates with PDF/Excel export
- Scheduled Reports with CRUD, frequency selection, and delivery history
- TOTP 2FA with QR code setup and 10 backup codes
- 5-step Onboarding Wizard for new users
- Audit Trail page with IP-logged action tracking
- Command Palette (Ctrl+K) for quick navigation

Milestone 4: Bug Fixes, Testing & Production Readiness

Sep 4 - Sep 5, 2026 | 31 commits

- Sales pagination fix - 7 consumer pages updated to handle paginated response
- Activity router rewrite - mapped to correct AuditLog model and columns
- Missing activity_router import in main.py fixed (was returning 404)
- 11 empty except blocks replaced with proper warning logs across 6 backend files
- 12 pages with missing PageSkeleton import fixed (crashed on load)
- CORS fix - added 127.0.0.1 origins alongside localhost
- Warm-up task fixed - corrected non-existent function name
- Dashboard Builder default template loads Executive layout on first visit
- Comprehensive testing: 60 API tests across 4 roles x 15 endpoints = 100% pass rate

Commit Statistics

Milestone	Commits	Date Range
Milestone 1: Foundation	22	Jul 24 - Aug 11
Milestone 2: AI/ML Integration	14	Aug 11 - Aug 28
Milestone 3: Platform Features	20	Sep 3 - Sep 4
Milestone 4: Bug Fixes & Testing	31	Sep 4 - Sep 5
Total	87	Jul 24 - Sep 5, 2026

12. Bug Fix Log (16 Critical Bugs)

#	Bug	Root Cause	Fix
1	JWT token lost on page refresh	Token not stored persistently	localStorage with auto-attach axios interceptor
2	Any user could access any page	No route-level role checks	ProtectedRoute with allowedRoles array
3	Neon connections dropping after 5 minutes	5-minute serverless idle timeout	pool_pre_ping=True + keepalive background task
4	AI pages crash on empty data	No null checks on API responses	Fallback catches + EmptyState component guards
5	Churn returns 0 predictions	ttl_cache caching empty cold-start results	Backend restart + cache invalidation
6	Too many anomaly false positives	Thresholds too loose (2 sigma)	Tightened to 2.5 sigma + business rule anomalies module
7	Merge conflicts from 4 branches	Overlapping main.py, App.jsx, requirements.txt	Manual resolution of 15+ conflicts
8	ChatBot gives useless answers	No knowledge base	Built RAG with business-specific Q&A pairs + TTS
9	i18n crashes entire app	Missing closing quote in Hindi translation	Fixed syntax; verified all 5 language blocks
10	Scheduled Reports not saving data	_get_bid() returns None; wrong ID format	Decode JWT directly; validate numeric IDs
11	Install modal won't open from homepage	CustomEvent not reliably received in iframe	Exposed window.__openInstallModal global function
12	Dashboard Builder empty on first load	No fallback when Neon returns empty layout	Auto-loads Executive DEFAULT_LAYOUT template
13	7 pages crash after sales pagination	sales.filter called on paginated object, not array	Array.isArray fallback in all 7 consumer pages
14	Activity router 500 error	Referenced models.Activity (doesn't exist)	Rewrote to use AuditLog with correct column mapping
15	Role logins failing	password_hash vs hashed_password field mismatch	Reset all passwords using correct hashed_password field
16	CORS blocking browser requests	127.0.0.1 missing from CORS_ORIGINS	Added 127.0.0.1:5173 and 127.0.0.1:5174 variants

13. Key Learnings

Multi-tenant architecture

Every query must filter by business_id. One missed filter = data leak between businesses.

Backend changes ripple

Changing sales response from array to paginated object broke 7 frontend pages instantly.

Merge conflicts are inevitable

4 developers need clear file ownership and frequent small commits to avoid conflicts.

Error handling is mandatory

11 empty except blocks hid real bugs silently for weeks.

CORS is browser-specific

localhost and 127.0.0.1 are treated as different origins by browsers.

ML needs real data

Synthetic training data produces perfect metrics but meaningless real-world predictions.

PWA requires HTTPS

Install prompts only trigger on HTTPS or localhost - not on LAN IP addresses.

ORM field names matter

password_hash vs hashed_password caused silent SQLAlchemy write failures.

14. Getting Started

Prerequisites

- Node.js 18+ and npm
- Python 3.9+
- Neon account (free tier) for PostgreSQL
- Git

Backend Setup

```
$ git clone https://github.com/springboardmentor24052s-tech/Team_2_Small_Biz_AI.git
$ cd Team_2_Small_Biz_AI/backend
$ python -m venv venv
$ venv\Scripts\activate      # Windows
$ pip install -r requirements.txt
$ cp .env.example .env       # Edit with your Neon DATABASE_URL
$ uvicorn app.main:app --reload --port 8000
```

Backend: <http://localhost:8000> | API Docs: <http://localhost:8000/docs>

Required Environment Variables

Variable	Required	Description
DATABASE_URL	Yes	Neon PostgreSQL connection string (pooler endpoint)
JWT_SECRET_KEY	Recommend	Random secret for JWT tokens (auto-generated if empty)
CORS_ORIGINS	No	Comma-separated frontend URLs (defaults to localhost)
SENDER_EMAIL	Optional	Gmail for OTP password reset emails
SENDER_PASSWORD	Optional	Gmail app password for OTP emails

Frontend Setup

```
$ cd frontend
$ npm install
$ npm run dev
```

Frontend: <http://localhost:5173>

First Login

- Open <http://localhost:5173>
- Click Sign Up and create an account with role: Business Owner
- Complete the 5-step onboarding wizard
- Start adding products, sales, and customers
- Explore AI insights from the sidebar navigation

MarketMind AI

Making enterprise-grade analytics accessible to every shop owner

Built for Indian Small Businesses

Infosys Springboard Capstone Project - Team 2

Neelam Rishika Damini | 2026

This project is for educational purposes as part of the Infosys Springboard program.